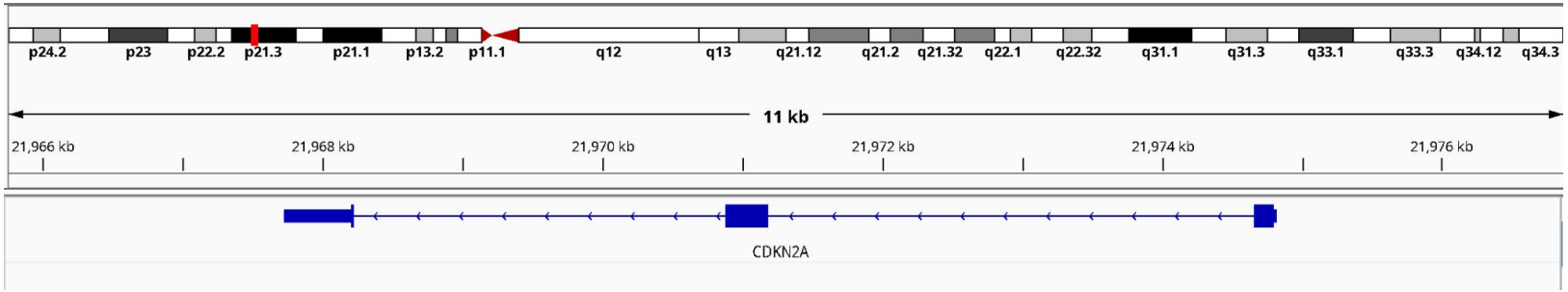
Role of CNVs in CDKN2A on cancer in eye, brain stem, and meninges

Anais, Martin

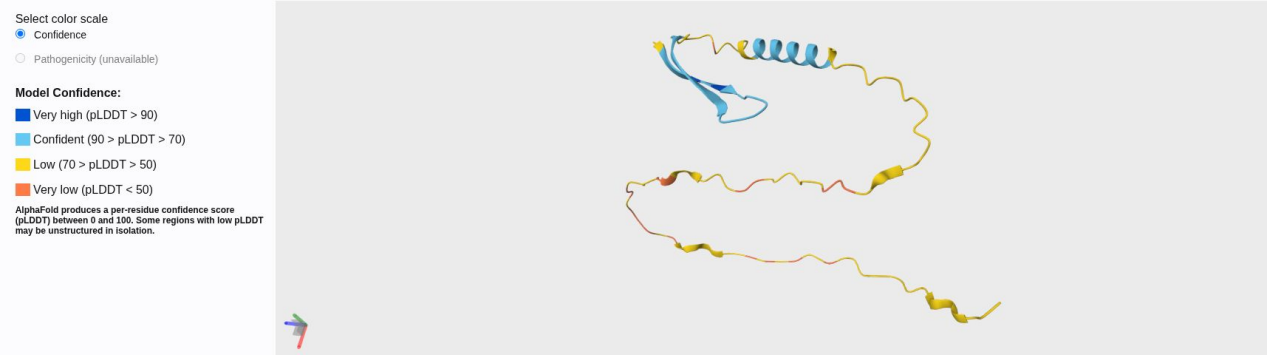
Gene: CDKN2A


p24.2
p23
p22.2
p21.3
p21.1
p13.2
p11.1
q12
q13
q21.12
q21.2
q21.32
q22.1
q22.32
q31.1
q31.3
q33.1
q33.3
q34.12
q34.3
11 kb
21,966 kb
21,968 kb
21,970 kb
21,972 kb
21,974 kb
21,976 kb
CDKN2A

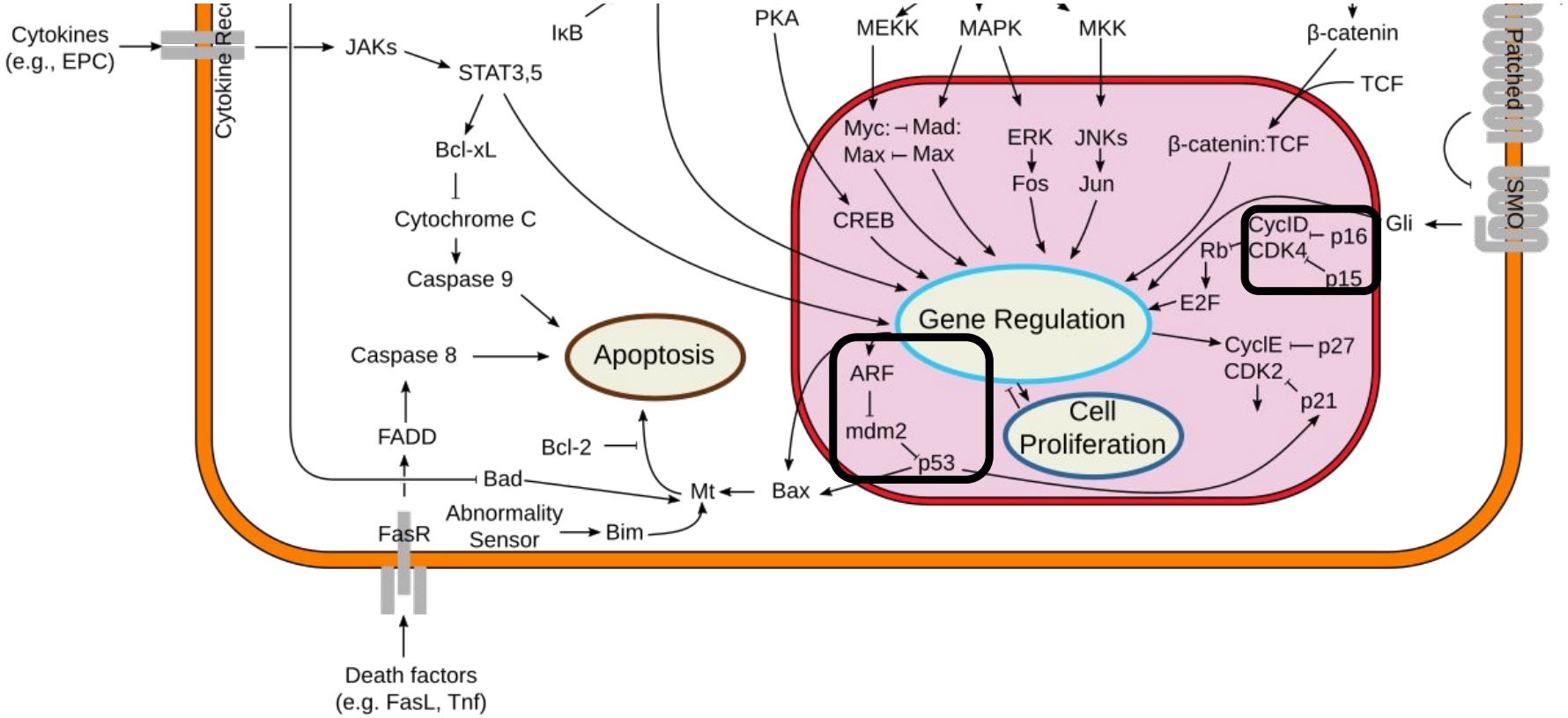
Solution NMR structure of p16

Multiple important isoforms:

- CDK inhibitors (particularly CDK4) and cell cycle G1 control, for example p16
- ARF product - stabilization of p53 degradation via interaction with p53-degrading E3 ubiquitin-protein ligase Mdm2



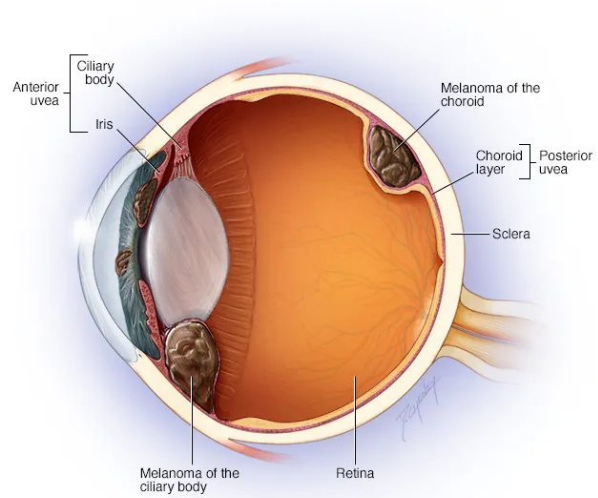
Predicted structure of ARF product (AlphaFold DB)



Overview of signal transduction pathways involved in apoptosis.
(https://en.wikipedia.org/wiki/Cyclin-dependent_kinase_4#/media/File:Signal_transduction_pathways.svg)

Background on examined cancer types

Eye melanoma



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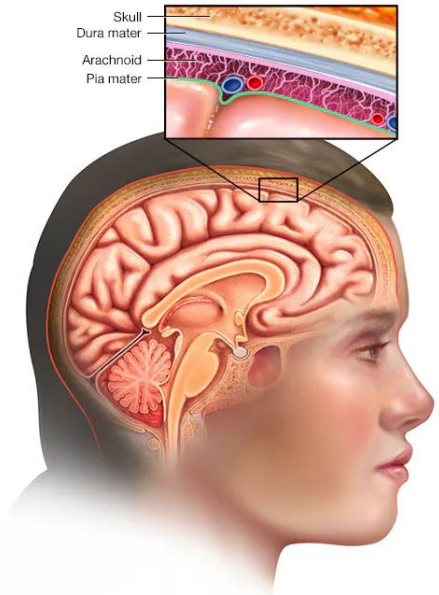
Meningioma



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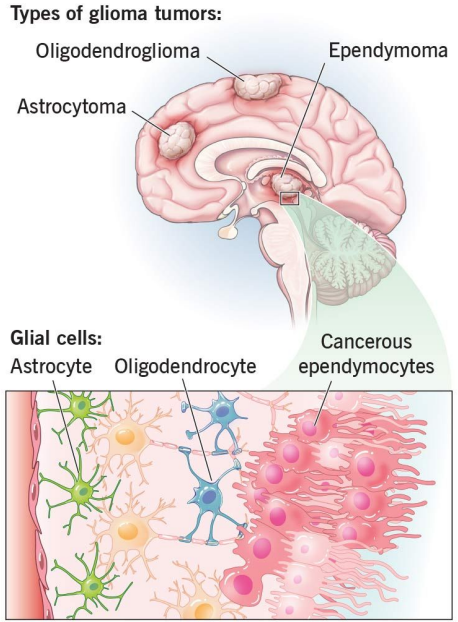
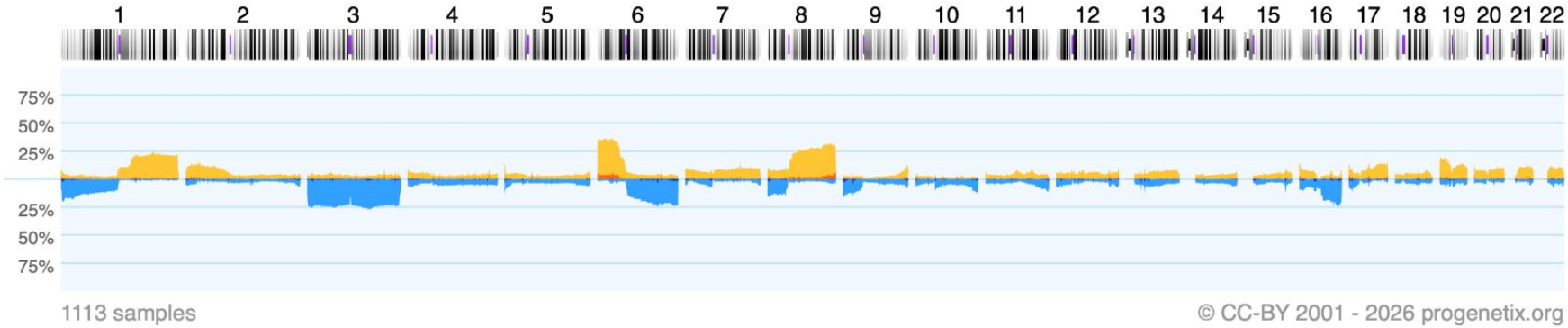
Glioma



Cleveland Clinic ©2025

Subset Type 1: Eye, Nos



- Nos “Not Otherwise Specified”
- 1113 samples
- After filtering for usable survival data: 181 individuals

Results for CDKN2A - eye, nos

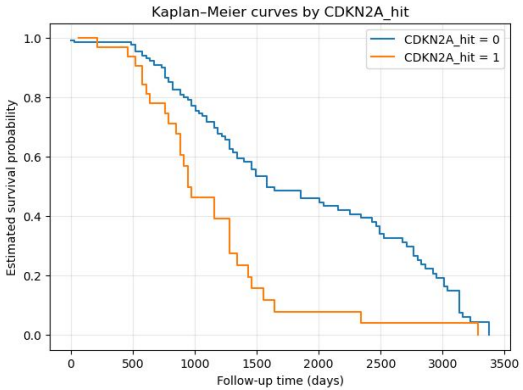
CNVs

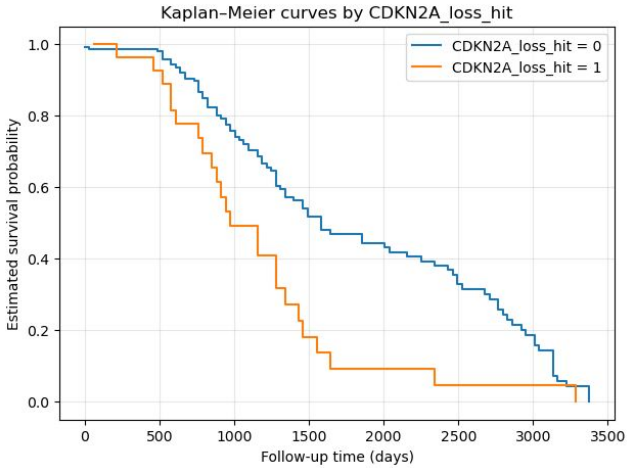
- **Gain hit:** 16/181 (≈ 9%)
- **Loss hit:** 7/181 (≈ 4%)
- Losses were rare, while gains were more common

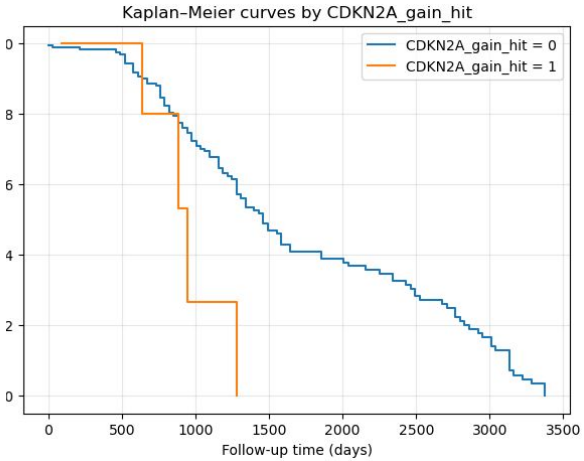
Survival Impact (Kaplan-Meier curves)

- **Hit: worse survival** (steep early decline) vs. the no-hit group
- **Loss hit: strong negative effect** on survival, with rapid drop-off
- **Gain hit:** Gain also associates with **poorer prognosis**, though the separation is somewhat less dramatic than loss

	hit	Loss hit	Gain hit
0	146	174	153
1	35	7	16







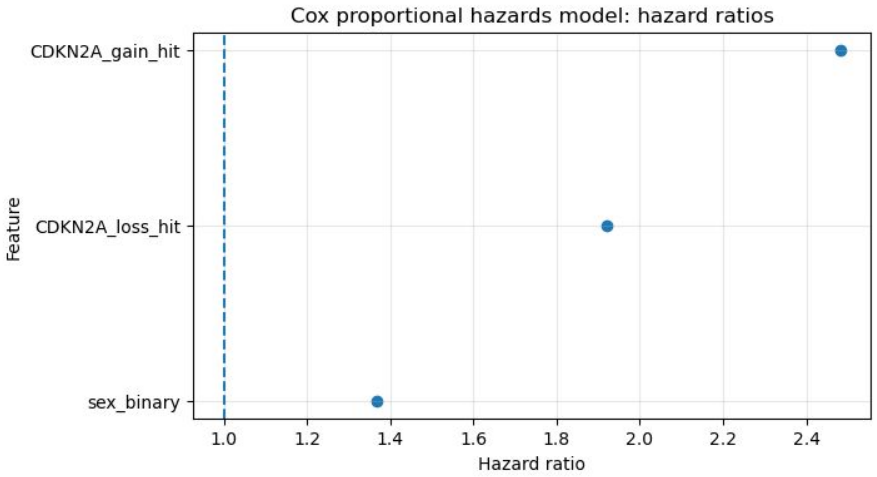
Results for CDKN2A - eye, nos

CNVs

- **Gain hit:** 16/181 (≈ 9%)
- **Loss hit:** 7/181 (≈ 4%)
- Losses were rare, while gains were more common

Cox Hazard model

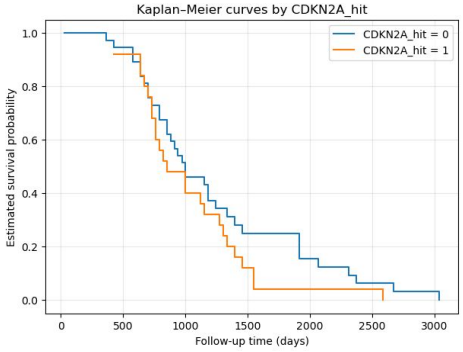
- **Loss hit and gain hit: worse survival** - high hazard ratio
- **Sex:** Male sex moderately higher hazard ratio, showing male patients are associated with lower survival rate

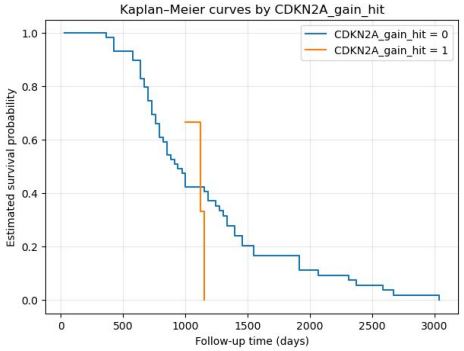


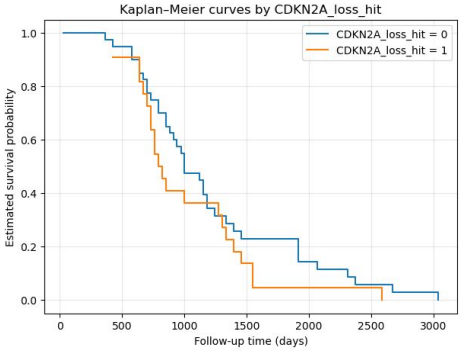
Feature	Gain hit	Loss hit	Sex
Coefficient	0.91	0.65	0.31

Concordance index: 0.57

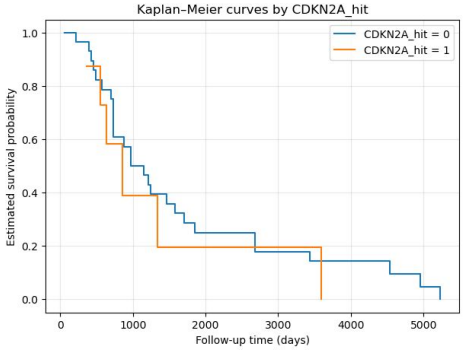
Results for CDKN2A

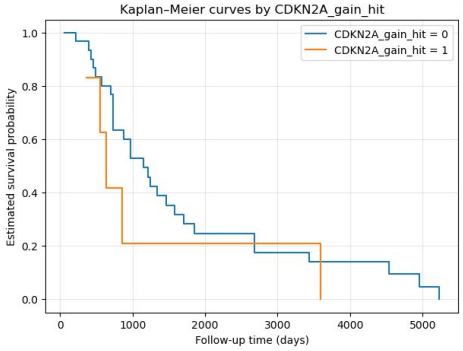


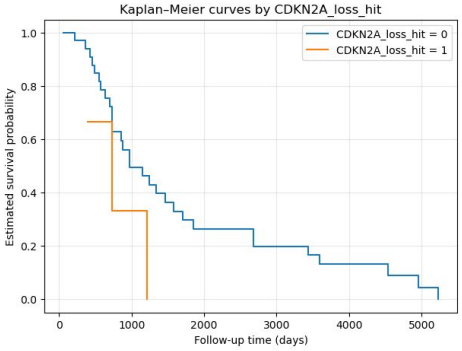




Meninges cohort (63 individuals)

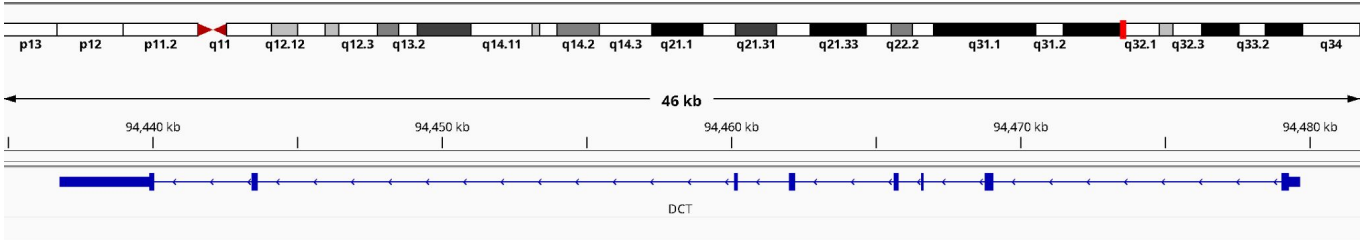






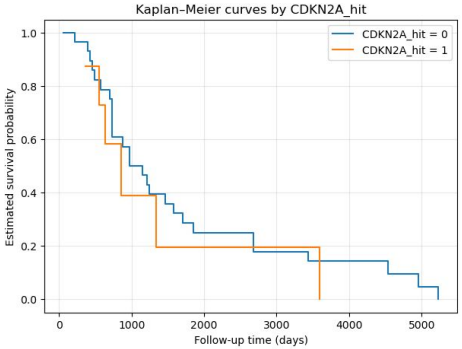
Brain stem cohort (40 individuals)

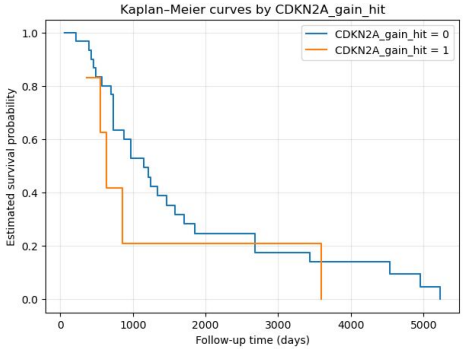
Gene: DCT (control)



- Codes for L-Dopachrome Tautomerase
- Role in melanin biosynthesis
- Not associated with carcinogenesis

Brain stem


Kaplan–Meier curves by CDKN2A_hit
CDKN2A_hit = 0
CDKN2A_hit = 1
Estimated survival probability
Follow-up time (days)


Kaplan–Meier curves by CDKN2A_gain_hit
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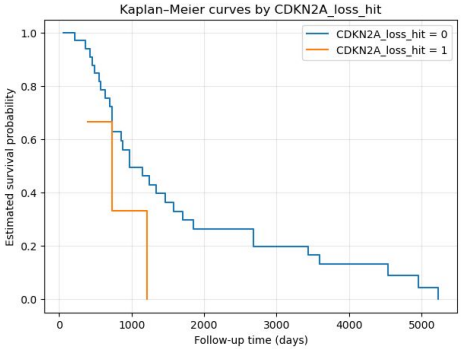
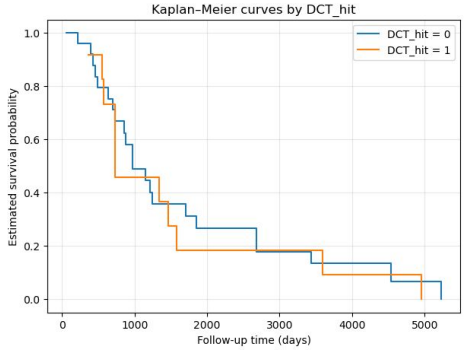
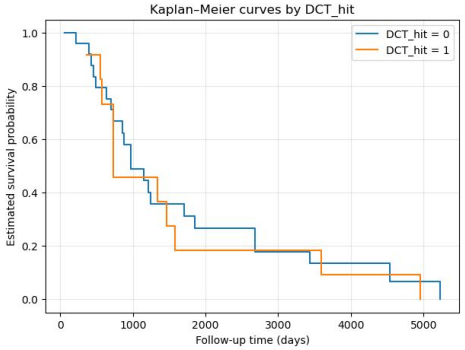
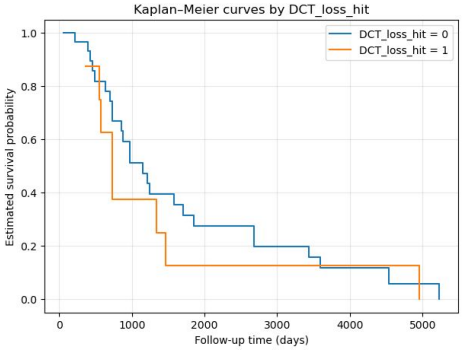
CDKN2A_gain_hit = 1
Estimated survival probability
Follow-up time (days)

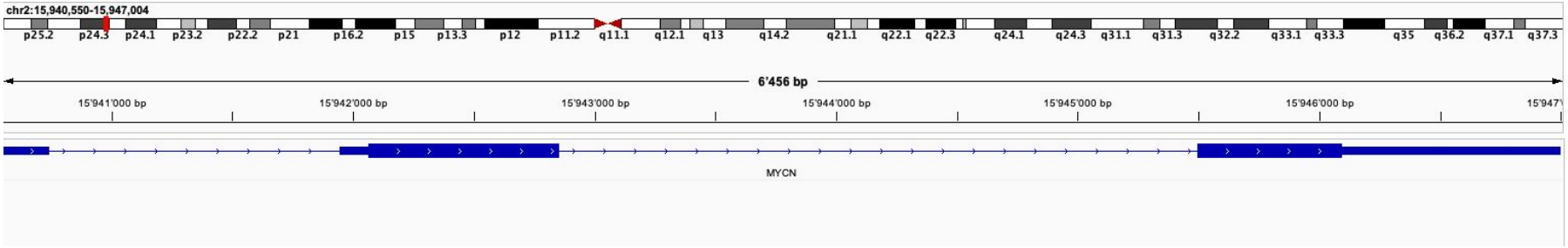

Kaplan–Meier curves by CDKN2A_loss_hit
CDKN2A_loss_hit = 0
CDKN2A_loss_hit = 1
Estimated survival probability
Follow-up time (days)


Kaplan–Meier curves by DCT_hit
DCT_hit = 0
DCT_hit = 1
Estimated survival probability
Follow-up time (days)


Kaplan–Meier curves by DCT_hit
DCT_hit = 0
DCT_hit = 1
Estimated survival probability
Follow-up time (days)


Kaplan–Meier curves by DCT_loss_hit
DCT_loss_hit = 0
DCT_loss_hit = 1
Estimated survival probability
Follow-up time (days)

Gene: MYCN


chr2:15,940,550-15,947,004
p25.2 p24.3 p24.1 p23.2 p22.2 p21 p16.2 p15 p13.3 p12 p11.2 q11.1 q12.1 q13 q14.2 q21.1 q22.1 q22.3 q24.1 q24.3 q31.1 q31.3 q32.2 q33.1 q33.3 q35 q36.2 q37.1 q37.3
6'456 bp
15'941'000 bp
15'942'000 bp
15'943'000 bp
15'944'000 bp
15'945'000 bp
15'946'000 bp
MYCN

MYCN Overview

- Chr 2
- Protein coding gene (N-Myc)
- 7 transcripts, 7 proteins
- Member of the MYC gene family (Transcriptions Factors)
- Protein contains a basic helix-loop-helix (bHLH) domain
- Onkogene
  - amplification/overexpression can lead to tumorigenesis

Results for MYCN - brain stem

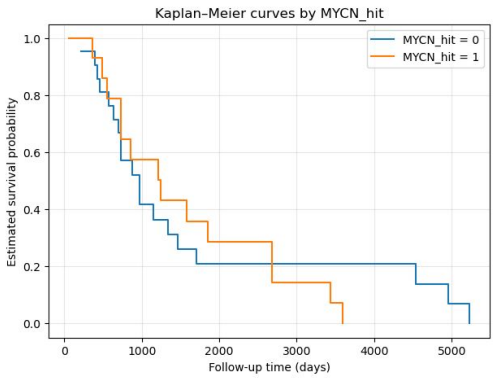
CNVs

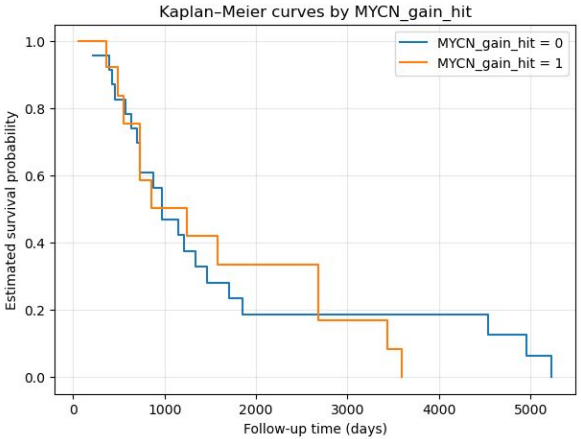
- **Gain Hit:** 16/40 (≈ 40%)
- **Loss:** 2/40 (≈ 5%)
- Losses were rare, while gains were more common

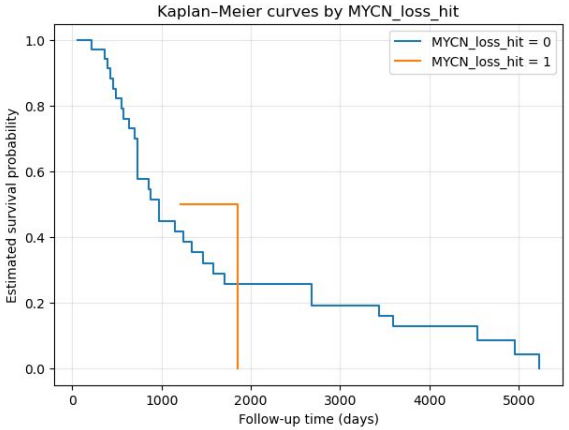
Survival Impact (Kaplan-Meier curves)

- **Hit:** initially similar, after 3500 days **worse survival** (steep early decline) vs. the no-hit group
- **Loss hit: strong negative effect** on survival, with rapid drop-off
- **Gain hit:** Gain also associates with **poorer prognosis**

	hit	Loss hit	Gain hit
0	22	38	24
1	18	2	16







Results for MYCN - brain stem

CNVs

- **Hit:** 35/181 (≈19%)
- **Loss:** 7/174 (≈ 4%)
- Losses were rare, while gains were more common

Cox Hazard model

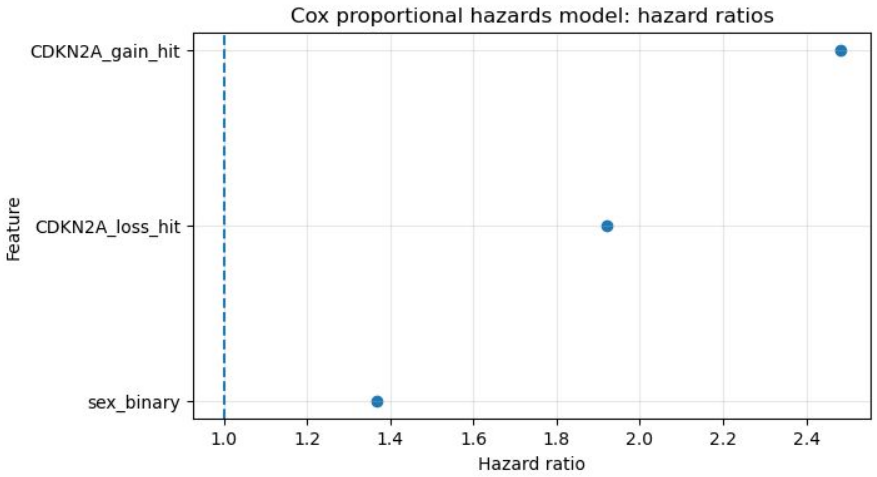
All 3 features have HR > 1 (dashed line), meaning **all are associated with increased mortality risk**.

Hazard ratio

- **Gain hit:** ~16% **lower** risk of death vs. no gain
- **Loss hit:** ~36% **lower** risk of death vs. no loss
- **Sex:** ~88% **higher** risk depending on sex

Concordance index = 0.60:

→ slightly better than random chance (0.50), but **weak predictive power** overall.



Feature	Gain hit	Loss hit	Sex
Hazard ratio	0.84	0.64	1.88

Concordance index: 0.6

Conclusions and Discussion

- CDKN2A alterations - both gains and losses - positively correlate with lower survival across tissues and cancer types
- Male sex also higher hazard association in eye and brain stem
- Low concordance index (<60) suggests either genes not as useful as biomarkers or insufficient sample size